


```

:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
{FUNCEXP
:funcid 1740
:funcresulttype 1700
:funcresultset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 0 0 0 0 0 0 0 0 ]
}
)
:location -1
}
)
:location -1
}
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
{RANGETBENTRY
:alias
{ALIAS
:aliasname n
:colnames <>
}
:eref
{ALIAS
:aliasname n
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_
comment")
}
:rtekind 0
:relid 17158
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname r
:colnames <>
}
:eref
{ALIAS
:aliasname r
:colnames ("r_regionkey" "r_name" "r_comment")
}
}
:rtekind 0
:relid 17164
:inh true
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17158
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17164
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBENTRY
:rtindex 1
}
{RANGETBENTRY
:rtindex 2
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:varstype 23

```

```

        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 1
        :location -1
    }
    {VAR
        :varno 1
        :varattno 4
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 1
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 4
        :location -1
    }
}
:location -1
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
    {VAR
        :varno 1
        :varattno 3
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 3
        :location -1
    }
    {VAR
        :varno 2
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 2
        :varattnosyn 1
        :location -1
    }
)
:location -1
}
{OPEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
    {VAR
        :varno 2
        :varattno 2
        :vartype 1042
        :vartypmod 29
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 2
        :varattnosyn 2
        :location -1
    }
    {CONST
        :consttype 1042
        :consttypmod -1
        :constcollid 100
        :constlen -1
        :constbyval false
        :constisnull false
        :location -1
        :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
    }
)
:location -1
}
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
    {TARGETENTRY
        :expr
        {CONST
            :consttype 23
            :consttypmod -1
            :constcollid 0
            :constlen 4
            :constbyval true
            :constisnull false
            :location -1
            :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
        }
        :resno 1
        :resname ?column?
        :ressortgroupref 0
        :resorigtbl 0
        :resorigcol 0
        :resjunk false
    }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}

```

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    )
    :location -1
  }
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 1
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 1
      :location -1
    }
    :resno 1
    :resname s_supkey
    :ressortgroupref 0
    :resorigtbl 17181
    :resorigcol 1
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 2
      :vartype 1042
      :vartypmod 29
      :varcollid 100
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 2
      :location -1
    }
    :resno 2
    :resname s_name
    :ressortgroupref 0
    :resorigtbl 17181
    :resorigcol 2
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 3
      :vartype 1043
      :vartypmod 44
      :varcollid 100
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 3
      :location -1
    }
    :resno 3
    :resname s_address
    :ressortgroupref 0
    :resorigtbl 17181
    :resorigcol 3
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 4
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 4
      :location -1
    }
    :resno 4
    :resname s_nationkey
    :ressortgroupref 0
    :resorigtbl 17181
    :resorigcol 4
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 5
      :vartype 1042
      :vartypmod 19
      :varcollid 100
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 5
      :location -1
    }
    :resno 5
    :resname s_phone
    :ressortgroupref 0
    :resorigtbl 17181
    :resorigcol 5
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 6
      :vartype 1700
      :vartypmod 983046
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 6
      :location -1
    }
    :resno 6
    :resname s_acctbal
    :ressortgroupref 0
    :resorigtbl 17181
    :resorigcol 6
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 7
      :vartype 1043
      :vartypmod 105
      :varcollid 100

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        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 7
        :location -1
    }
    :resno 7
    :resname s_comment
    :ressortgroupref 0
    :resorigtbl 17181
    :resorigcol 7
    :resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:security_barrier false
:relid 17263
:inh false
:relkind v
:rellockmode 1
:perminfoindex 2
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname partsupp
:colnames ("ps_partkey" "ps_suppkey" "ps_availqty" "ps_supplycost" "p
s_comment")
}
:rtekind 0
:relid 17191
:inh true
:relkind r
:rellockmode 1
:perminfoindex 3
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname nation
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
:rtekind 0
:relid 17158
:inh true
:relkind r
:rellockmode 1
:perminfoindex 4
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname region
:colnames ("r_regionkey" "r_name" "r_comment")
}
:rtekind 0
:relid 17164
:inh true
:relkind r
:rellockmode 1
:perminfoindex 5
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17169
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10 12 13)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17263
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17191
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17158
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17164
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)

```

```

    }
}
:jointree
{FROMEXPR
:fromlist (
  {RANGETBLREF
  :rtindex 1
  }
  {RANGETBLREF
  :rtindex 2
  }
  {RANGETBLREF
  :rtindex 3
  }
  {RANGETBLREF
  :rtindex 4
  }
  {RANGETBLREF
  :rtindex 5
  }
)
:quals
{BOOLEXP
:boolop and
:args (
  {OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 290
}
  {VAR
:varno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 302
}
)
:location 300
}
  {OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location 325
}
  {VAR
:varno 3
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 2
:location 337
}
)
:location 335
}
  {OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
:varno 1
:varattno 6
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 360
}
  {CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 369
:constvalue 4 [ 15 0 0 0 0 0 0 0 ]
}
)
:location 367
}
  {OPEXP
:opno 1289
:opfuncid 850
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
  {RELABELTYPE
:arg
  {VAR
:varno 1
:varattno 5
:vartype 1043
:vartypmod 29

```

```

        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 5
        :location 384
    }
    :resulttype 25
    :resulttypemod -1
    :resultcollid 100
    :relabelformat 2
    :location -1
}
{CONST
:consttype 25
:consttypemod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 396
:constvalue 10 [ 40 0 0 0 37 66 82 65 83 83 ]
}
}
:location 391
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 4
:vartype 23
:vartypemod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 4
:location 417
}
{VAR
:varno 4
:varattno 1
:vartype 23
:vartypemod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location 431
}
)
:location 429
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 4
:varattno 3
:vartype 23
:vartypemod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 3
:location 455
}
{VAR
:varno 5
:varattno 1
:vartype 23
:vartypemod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 1
:location 469
}
)
:location 467
}
{OPEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 5
:varattno 2
:vartype 1042
:vartypemod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 2
:location 500
}
{CONST
:consttype 1042
:consttypemod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 516
:constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
}
)
:location 514
}
{OPEXPR
:opno 1752
:opfuncid 1718
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 4
:vartype 1700
:vartypemod 983046

```

```

:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 4
:location 537
}
{SUBLINK
:subLinkType 4
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname partsupp
:colnames ("ps_partkey" "ps_suppkey" "ps_availqty" "ps_supplycost" "ps_comment")
}
:rtekind 0
:relid 17191
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname supplier_secure
:colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "s_acctbal" "s_comment")
}
:rtekind 1
:subquery
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTY
:alias
{ALIAS
:aliasname s
:colnames <>
}
:eref
{ALIAS
:aliasname s
:colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "s_acctbal" "s_comment")
}
:rtekind 0
:relid 17181
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperinfos (
{RTEPERMISSIONINFO
:relid 17181
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12 13 14)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
)
}
)
}
}

```

```

{FUNCEXP
:funcid 1740
:funcresulttype 1700
:funcreset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 0 0 0 0 0 0 0 ]
}
)
:location -1
}
:location -1
}
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
{RANGETBENTRY
:alias
{ALIAS
:aliasname n
:colnames <>
}
:eref
{ALIAS
:aliasname n
:colnames ("n_nationkey" "n_name" "n_
regionkey" "n_comment")
}
:rtekind 0
:relid 17158
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname r
:colnames <>
}
:eref
{ALIAS
:aliasname r
:colnames ("r_regionkey" "r_name" "r_
comment")
}
:rtekind 0
:relid 17164
:inh true
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
:rteperinfos (
{RTEPERMISSIONINFO
:relid 17158
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17164
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
}
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
)
}
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opreset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
}
)
}
)
}

```

```

        :varnosyn 1
        :varattnosyn 1
        :location -1
    }
    {VAR
    :varno 1
    :varattno 4
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 1
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 4
    :location -1
    }
    )
    :location -1
    }
    {OPEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
    {VAR
    :varno 1
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 3
    :location -1
    }
    {VAR
    :varno 2
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 1
    :location -1
    }
    )
    :location -1
    }
    {OPEXPR
    :opno 1054
    :opfuncid 1048
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 100
    :args (
    {VAR
    :varno 2
    :varattno 2
    :vartype 1042
    :vartypmod 29
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 2
    :location -1
    }
    {CONST
    :consttype 1042
    :consttypmod -1
    :constcollid 100
    :constlen -1
    :constbyval false
    :constisnull false
    :location -1
    :constvalue 10 [ 40 0 0 0 69 8
    5 82 79 80 69 ]
    }
    )
    :location -1
    }
    )
    :location -1
    }
    }
    :mergeActionList <>
    :mergeTargetRelation 0
    :mergeJoinCondition <>
    :targetList (
    {TARGETENTRY
    :expr
    {CONST
    :consttype 23
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location -1
    :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
    }
    :resno 1
    :resname ?column?
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
    }
    )
    :override 0
    :onConflict <>
    :returningOldAlias <>
    :returningNewAlias <>
    :returningList <>
    :groupClause <>
    :groupDistinct false
    :groupingSets <>
    :havingQual <>
    :windowClause <>
    :distinctClause <>
    :sortClause <>
    :limitOffset <>
    :limitCount <>
    :limitOption 0
    :rowMarks <>
    :setOperations <>
    :constraintDeps <>
    :withCheckOptions <>
    :stmt_location -1
    :stmt_len -1
    }
    :location -1
    }
    )
    :location -1
    }
}

```

```

:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:varstype 23
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname s_supplekey
:ressortgroupref 0
:resorigtbl 17181
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:varstype 1042
:varstypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
:resno 2
:resname s_name
:ressortgroupref 0
:resorigtbl 17181
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:varstype 1043
:varstypmod 44
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
:resno 3
:resname s_address
:ressortgroupref 0
:resorigtbl 17181
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:varstype 23
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
:resno 4
:resname s_nationkey
:ressortgroupref 0
:resorigtbl 17181
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:varstype 1042
:varstypmod 19
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location -1
}
:resno 5
:resname s_phone
:ressortgroupref 0
:resorigtbl 17181
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:varstype 1700
:varstypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 6
:resname s_acctbal
:ressortgroupref 0
:resorigtbl 17181
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:varstype 1043
:varstypmod 105
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1

```

```

        :varattnosyn 7
        :location -1
    }
    :resno 7
    :resname s_comment
    :ressortgroupref 0
    :resorigtbl 17181
    :resorigcol 7
    :resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:security_barrier false
:relid 17263
:inh false
:relkind v
:rellockmode 1
:perminfoindex 2
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname nation
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_com
ment")
}
:rtekind 0
:relid 17158
:inh true
:relkind r
:rellockmode 1
:perminfoindex 3
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname region
:colnames ("r_regionkey" "r_name" "r_comment")
}
:rtekind 0
:relid 17164
:inh true
:relkind r
:rellockmode 1
:perminfoindex 4
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17191
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17263
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17158
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17164
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
{RANGETBLREF
:rtindex 3
}
{RANGETBLREF
:rtindex 4
}
)
)
:quals
{BOOLEXP
:boolop and
:args (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR

```

```

:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 826
}
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 838
}
}
:location 836
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location 877
}
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 889
}
}
}
:location 887
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 4
:location 928
}
{VAR
:varno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 942
}
}
}
:location 940
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 3
:location 982
}
{VAR
:varno 4
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location 996
}
}
}
:location 994
}
{OPEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR

```

```

        :varno 4
        :varattno 2
        :vartype 1042
        :vartypmod 29
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 4
        :varattnosyn 2
        :location 1043
    }
    {CONST
    :consttype 1042
    :consttypmod -1
    :constcollid 100
    :constlen -1
    :constbyval false
    :constisnull false
    :location 1059
    :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
    }
    }
    :location 1057
    }
    }
    :location 873
    }
    }
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetlist (
    {TARGETENTRY
    :expr
    {AGGREF
    :aggnoid 2146
    :aggtype 1700
    :aggcollid 0
    :inputcollid 0
    :aggtranstype 0
    :aggargtypes (o 1700)
    :aggdirectargs <>
    :args (
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 4
    :vartype 1700
    :vartypmod 983046
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 4
    :location 606
    }
    :resno 1
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
    }
    )
    :aggorder <>
    :aggdistinct <>
    :aggfilter <>
    :aggstar false
    :aggvariadic false
    :aggkind n
    :aggpresorted false
    :agglevelsup 0
    :aggssplit 0
    :aggno -1
    :aggtransno -1
    :location 602
    }
    :resno 1
    :resname min
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
    }
    )
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
    }
    :location 553
    }
    }
    :location 551
    }
    }
    :location 321
    }
    }
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetlist (
    {TARGETENTRY
    :expr
    {VAR
    :varno 2
    :varattno 6
    :vartype 1700
    :vartypmod 983046
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 6
    :location 51
    }
    :resno 1
    :resname s_acctbal
    :ressortgroupref 1
    :resorigtbl 17263
    :resorigcol 6
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR

```

```

:varno 2
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location 70
}
:resno 2
:resname s_name
:ressortgroupref 3
:resorigtbl 17263
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 2
:location 86
}
:resno 3
:resname n_name
:ressortgroupref 2
:resorigtbl 17158
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 102
}
:resno 4
:resname p_partkey
:ressortgroupref 4
:resorigtbl 17169
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 121
}
:resno 5
:resname p_mfgr
:ressortgroupref 0
:resorigtbl 17169
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 3
:vartype 1043
:vartypmod 44
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 3
:location 137
}
:resno 6
:resname s_address
:ressortgroupref 0
:resorigtbl 17263
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 5
:vartype 1042
:vartypmod 19
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 5
:location 156
}
:resno 7
:resname s_phone
:ressortgroupref 0
:resorigtbl 17263
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 7
:vartype 1043
:vartypmod 105
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 7
:location 173
}
:resno 8
:resname s_comment
:ressortgroupref 0
:resorigtbl 17263

```

```

:resorigcol 7
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
  {SORTGROUPCLAUSE
    :tleSortGroupRef 1
    :eqop 1752
    :sortop 1756
    :reverse_sort true
    :nulls_first true
    :hashable true
  }
  {SORTGROUPCLAUSE
    :tleSortGroupRef 2
    :eqop 1054
    :sortop 1058
    :reverse_sort false
    :nulls_first false
    :hashable true
  }
  {SORTGROUPCLAUSE
    :tleSortGroupRef 3
    :eqop 1054
    :sortop 1058
    :reverse_sort false
    :nulls_first false
    :hashable true
  }
  {SORTGROUPCLAUSE
    :tleSortGroupRef 4
    :eqop 96
    :sortop 97
    :reverse_sort false
    :nulls_first false
    :hashable true
  }
)
:limitOffset <>
:limitCount
{FUNCEXP
  :funcid 481
  :funcresulttype 20
  :funcresultset false
  :funcvariadic false
  :funcformat 2
  :funccollid 0
  :inputcollid 0
  :args (
    {CONST
      :consttype 23
      :consttypmod -1
      :constcollid 0
      :constlen 4
      :constbyval true
      :constisnull false
      :location 1167
      :constvalue 4 [ 100 0 0 0 0 0 0 0 ]
    }
  )
  :location -1
}
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

WARNING: max_parallel_hazard_walker: FuncExpr funcid=1740 proparallel=s (s=safe r=restricted u=unsafe)
WARNING: max_parallel_hazard_walker: FuncExpr funcid=481 proparallel=s (s=safe r=restricted u=unsafe)
WARNING: max_parallel_hazard_walker: FuncExpr funcid=1740 proparallel=s (s=safe r=restricted u=unsafe)
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctlist <>
:rttable (
  {RANGETBENTRY
    :alias <>
    :eref
      {ALIAS
        :aliasname partsupp
        :colnames ("ps_partkey" "ps_suppkey" "ps_availqty" "ps_supplycost" "p
          s_comment")
      }
    :rtekind 0
    :relid 17191
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
  {RANGETBENTRY
    :alias <>
    :eref
      {ALIAS
        :aliasname supplier_secure
        :colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "
          s_acctbal" "s_comment")
      }
    :rtekind 1
    :subquery <>
    :security_barrier false
    :relid 17263
    :inh false
    :relkind v
    :rellockmode 1
    :perminfoindex 2
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
  {RANGETBENTRY
    :alias <>
    :eref
      {ALIAS

```

```

        :aliasname nation
        :colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
    }
}
:rtekind 0
:relid 17158
:inh false
:relkind r
:rellockmode 1
:perminfoindex 3
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname region
:colnames ("r_regionkey" "r_name" "r_comment")
}
:rtekind 0
:relid 17164
:inh false
:relkind r
:rellockmode 1
:perminfoindex 4
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname s
:colnames <>
}
:eref
{ALIAS
:aliasname s
:colnames ("s_suppkkey" "s_name" "s_address" "s_nationkey" "s_phone" "
s_acctbal" "s_comment")
}
:rtekind 0
:relid 17181
:inh false
:relkind r
:rellockmode 1
:perminfoindex 5
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname n
:colnames <>
}
:eref
{ALIAS
:aliasname n
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
:rtekind 0
:relid 17158
:inh false
:relkind r
:rellockmode 1
:perminfoindex 6
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname r
:colnames <>
}
:eref
{ALIAS
:aliasname r
:colnames ("r_regionkey" "r_name" "r_comment")
}
:rtekind 0
:relid 17164
:inh false
:relkind r
:rellockmode 1
:perminfoindex 7
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17191
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17263
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17158
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17164
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17181
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12 13 14)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17158

```

```

:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17164
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 5
}
)
:quals (
{OPEXPR
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 5
:varattno 6
:varatype 1700
:varitypm 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 6
:location -1
}
)
{CONST
:consttype 1700
:constitypm -1
:constcollid 0
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 6 [ 24 0 0 0 0 -128 ]
}
)
:location -1
}
)
}
:rarg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 6
}
{RANGETBLREF
:rtindex 7
}
)
:quals <>
}
:usingClause <>
:join_using_alias <>
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 6
:varattno 1
:varatype 23
:varitypm -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 1
:location -1
}
{VAR
:varno 5
:varattno 4
:varatype 23
:varitypm -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 4
:location -1
}
)
:location -1
}
)
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 6
:varattno 3
:varatype 23
:varitypm -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 3
:location -1
}
{VAR
:varno 7
:varattno 1
:varatype 23
:varitypm -1

```

```

        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 7
        :varattnosyn 1
        :location -1
    }
}
:location -1
}
{OPEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
    {VAR
        :varno 7
        :varattno 2
        :vartype 1042
        :vartypmod 29
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 7
        :varattnosyn 2
        :location -1
    }
    {CONST
        :consttype 1042
        :consttypmod -1
        :constcollid 100
        :constlen -1
        :constbyval false
        :constisnull false
        :location -1
        :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
    }
)
:location -1
}
)
:alias <>
:rtindex 0
}
{RANGETBLREF
:rtindex 3
}
{RANGETBLREF
:rtindex 4
}
)
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
    {PARAM
        :paramkind 1
        :paramid 0
        :paramtype 23
        :paramtypmod -1
        :paramcollid 0
        :location 826
    }
    {VAR
        :varno 1
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 1
        :location 838
    }
)
:location 836
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
    {VAR
        :varno 5
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 5
        :varattnosyn 1
        :location -1
    }
    {VAR
        :varno 1
        :varattno 2
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 2
        :location 889
    }
)
:location 887
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
    {VAR
        :varno 5
        :varattno 4
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 5
        :varattnosyn 4
        :location -1
    }
)
{VAR

```

```

:varno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 942
}
)
:location 940
}
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 3
:location 982
}
{VAR
:varno 4
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location 996
}
)
:location 994
}
{OEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 4
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 2
:location 1043
}
{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 1059
:constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
}
)
:location 1057
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{AGGREF
:aggfncid 2146
:aggtype 1700
:aggcollid 0
:inputcollid 0
:aggtranstype 0
:aggargtypes (o 1700)
:aggdirectargs <>
:args (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location 006
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar false
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggsplit 0
:aggno -1
:aggtransno -1
:location 002
}
)
:resno 1
:resname min
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0

```

```
WARNING: is_parallel_safe: INPUT qual = (<{RESTRICINFO :clause (OEXPR :opno 96 :ofuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((PARAM :paramkind 1 :paramid 0 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1) (VAR :varno 1 :varrtno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varrattnosyn 1 :location -1) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b 1) :right_relds (b 1) :orclause => :rinfo_serial 5 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_select -1 :outer_select -1 :mergerefamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_exp (PARAM :paramkind 1 :paramid 0 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1) :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 3 4 5 6 7))) :right_em (EQUIVALENCEMEMBER :em_exp (VAR :varno 1 :varrtno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varrattnosyn 1 :location -1) :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 3 4 5 6 7))) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0})
WARNING: is_parallel_safe: OUTPUT qual = <>
WARNING: is_parallel_safe: INPUT qual = (<{RESTRICINFO :clause (OEXPR :opno 96 :ofuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((PARAM :paramkind 1 :paramid 0 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1) (VAR :varno 1 :varrtno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varrattnosyn 1 :location -1) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b 1) :right_relds (b 1) :orclause => :rinfo_serial 5 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_select -1 :outer_select -1 :mergerefamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_exp (PARAM :paramkind 1 :paramid 0 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1) :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 3 4 5 6 7))) :right_em (EQUIVALENCEMEMBER :em_exp (VAR :varno 1 :varrtno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varrattnosyn 1 :location -1) :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 3 4 5 6 7))) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0}) | parallel_safe = false | reason = PROP
ROPARALLEL_RESTRICTED
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = (<{VAR :varno 3 :varrtno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varrattnosyn 1 :location -1) (VAR :varno 3 :varrtno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varrattnosyn 3 :location -1) (VAR :varno 3 :varrtno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varrattnosyn 3 :location -1) | parallel_safe = true | reason = PROP
ARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = (<{RESTRICINFO :clause (OEXPR :opno 1054 :ofuncid 1048 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ((VAR :varno 4 :varrtno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varrattnosyn 2 :location -1) (CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constnbval false :constnull false :location -1 :constvalue 10 [ 4 0 0 0 69 85 82 79 80 69 ]]) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 4) :required_relds (b 4) :incompatible_relds (b) :outer_relds (b 4) :right_relds (b 4) :orclause => :rinfo_serial 11 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_select -1 :outer_select -1 :mergerefamilies (o 426 2097) :left_em (EQUIVALENCEMEMBER :em_exp (VAR :varno 4 :varrtno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varrattnosyn 2 :location -1) :em_relds (b 4) :em_is_const false :em_is_child false :em_datatype 1042 :em_jdomain (JOINDOMAIN :jd_relds (b 1 3 4 5 6 7))) :right_em (EQUIVALENCEMEMBER :em_exp (CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constnbval false :constnull false :location -1 :constvalue 10 [ 4 0 0 0 69 85 82 79 80 69 ]]) :em_relds (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain (JOINDOMAIN :jd_relds (b 1 3 4 5 6 7))) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = (<{RESTRICINFO :clause (OEXPR :opno 1054 :ofuncid 1048 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ((VAR :varno 4 :varrtno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varrattnosyn 2 :location -1) (CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constnbval false :constnull false :location -1 :constvalue 10 [ 4 0 0 0 69 85 82 79 80 69 ]]) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 4) :required_relds (b 4) :incompatible_relds (b) :outer_relds (b 4) :right_relds (b 4) :orclause => :rinfo_serial 11 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_select -1 :outer_select -1 :mergerefamilies (o 426 2097) :left_em (EQUIVALENCEMEMBER :em_exp (VAR :varno 4 :varrtno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varrattnosyn 2 :location -1) :em_relds (b 4) :em_is_const false :em_is_child false :em_datatype 1042 :em_jdomain (JOINDOMAIN :jd_relds (b 1 3 4 5 6 7))) :right_em (EQUIVALENCEMEMBER :em_exp (CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constnbval false :constnull false :location -1 :constvalue 10 [ 4 0 0 0 69 85 82 79 80 69 ]]) :em_relds (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain (JOINDOMAIN :jd_relds (b 1 3 4 5 6 7))) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = (<{VAR :varno 4 :varrtno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varrattnosyn 1 :location -1) (VAR :varno 5 :varrtno 5 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 5 :varrattnosyn 4 :location -1) (VAR :varno 5 :varrtno 5 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 5 :varrattnosyn 5 :location -1) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: OUTPUT qual = (<{VAR :varno 4 :varrtno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varrattnosyn 1 :location -1) (VAR :varno 5 :varrtno 5 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 5 :varrattnosyn 4 :location -1) (VAR :varno 5 :varrtno 5 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 5 :varrattnosyn 5 :location -1) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = (<{RESTRICINFO :clause (OEXPR :opno 1756 :ofuncid 1720 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 5 :varrtno 6 :vartype 17 :vartypmod 29 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 5 :varrattnosyn 6 :location -1) (CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen -1 :constnbval false :constnull false :location -1 :constvalue 6 [ 2 4 0 0 0 -128 ]]) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 5) :required_relds (b 5) :incompatible_relds (b) :outer_relds (b 5) :right_relds (b 5) :orclause => :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_select -1 :outer_select -1 :mergerefamilies (o 426 2097) :left_em (EQUIVALENCEMEMBER :em_exp (VAR :varno 5 :varrtno 6 :vartype 17 :vartypmod 29 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 5 :varrattnosyn 6 :location -1) :em_relds (b 5) :em_is_const false :em_is_child false :em_datatype 17 :em_jdomain (JOINDOMAIN :jd_relds (b 1 3 4 5 6 7))) :right_em (EQUIVALENCEMEMBER :em_exp (CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen -1 :constnbval false :constnull false :location -1 :constvalue 6 [ 2 4 0 0 0 -128 ]]) :em_relds (b) :em_is_const true :em_is_child false :em_datatype 17 :em_jdomain (JOINDOMAIN :jd_re
```

[illegible]

```
vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varattnosyn 1 :location -1} {VAR :varno 4 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 1 :location -1} | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 3 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 3 :location -1} {VAR :varno 4 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 1 :location -1} :location -1} :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 3 4) :required_relds (b 3 4) :incompatible_relds (b) :outer_relds (b 3) :right_relds (b 4) :orclause <> :rinfo_serial 15 :eval_cost.startup 0 :eval_cost.per_tuple 0.0025 :norm_select 0.0058823529411764705 :outer_select -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 3 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 3 :location -1} :em_relds (b 3) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 3 4 5 6 7)}} :right_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 4 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 1 :location -1} :em_relds (b 4) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 3 4 5 6 7)}} :outer_is_left false :hashjoin_operator 96 :left_bucketsize 0.0058823529411764705 :right_bucketsize 1 :left_mcvfreq 0 :right_mcvfreq 0 :left_hashoperator 96 :right_hashoperator 96})
WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 3 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 3 :location -1} :em_relds (b 3) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 3 4 5 6 7)}} :right_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 4 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 1 :location -1} :em_relds (b 4) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 3 4 5 6 7)}} :outer_is_left false :hashjoin_operator 96 :left_bucketsize 0.0058823529411764705 :right_bucketsize 1 :left_mcvfreq 0 :right_mcvfreq 0 :left_hashoperator 96 :right_hashoperator 96}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 3 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 1 :location -1} {VAR :varno 4 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varattnosyn 1 :location -1} :location -1} | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 3 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 3 :location -1} {VAR :varno 4 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 1 :location -1} :location -1} :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 3 4) :required_relds (b 3 4) :incompatible_relds (b) :outer_relds (b 3) :right_relds (b 4) :orclause <> :rinfo_serial 15 :eval_cost.startup 0 :eval_cost.per_tuple 0.0025 :norm_select 0.0058823529411764705 :outer_select -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 3 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 3 :location -1} :em_relds (b 3) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 3 4 5 6 7)}} :right_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 4 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 1 :location -1} :em_relds (b 4) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 3 4 5 6 7)}} :outer_is_left true :hashjoin_operator 96 :left_bucketsize 0.0058823529411764705 :right_bucketsize 1 :left_mcvfreq 0 :right_mcvfreq 0 :left_hashoperator 96 :right_hashoperator 96})
WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 3 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 3 :location -1} {VAR :varno 4 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 1 :location -1} :location -1} :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 3 4) :required_relds (b 3 4) :incompatible_relds (b) :outer_relds (b 3) :right_relds (b 4) :orclause <> :rinfo_serial 15 :eval_cost.startup 0 :eval_cost.per_tuple 0.0025 :norm_select 0.0058823529411764705 :outer_select -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 3 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 3 :location -1} :em_relds (b 3) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 3 4 5 6 7)}} :right_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 4 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 1 :location -1} :em_relds (b 4) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 3 4 5 6 7)}} :outer_is_left true :hashjoin_operator 96 :left_bucketsize 0.0058823529411764705 :right_bucketsize 1 :left_mcvfreq 0 :right_mcvfreq 0 :left_hashoperator 96 :right_hashoperator 96}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 5 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 5 :varattnosyn 1 :location -1} :location -1} | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((AGGREF :aggfnoid 2146 :aggtype 1700 :aggcollid 0 :inputcollid 0 :aggtranstype 1700 :aggargtypes (o 1700) :aggdirectargs <> :args ({TARGETENTRY :expr {VAR :varno 1 :varattno 4 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1} :resno 1 :resname <> :ressortgroupref 0 :resorigtbl 0 :resorigcol 0 :resjunk false}) :aggorder <> :aggdistinct <> :aggfilter <> :aggstar false :aggvariadic false :aggkind n :aggpresorted false :agglevelsup 0 :aggssplit 0 :aggno 0 :aggtransno 0 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ((AGGREF :aggfnoid 2146 :aggtype 1700 :aggcollid 0 :inputcollid 0 :aggtranstype 1700 :aggargtypes (o 1700) :aggdirectargs <> :args ({TARGETENTRY :expr {VAR :varno 1 :varattno 4 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1} :resno 1 :resname <> :ressortgroupref 0 :resorigtbl 0 :resorigcol 0 :resjunk false}) :aggorder <> :aggdistinct <> :aggfilter <> :aggstar false :aggvariadic false :aggkind n :aggpresorted false :agglevelsup 0 :aggssplit 0 :aggno 0 :aggtransno 0 :location -1}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 1 :varattno 4 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1} | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTe false
:isReturn false
:isList <>
:isTable (
:RANGELEFTENTRY
:alias <>
:eref
{ALIAS
:aliasname supplier_secure
:colnames ("s_supplekey" "s_name" "s_address" "s_nationkey" "s_phone" "s_acctbal" "s_comment")
}
:rtckind 0
:reldid 17169
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
:RANGELEFTENTRY
:alias <>
:eref
{ALIAS
:aliasname nation
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
:rtckind 0
:reldid 17158
```

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:inh false
:relkind r
:rellockmode 1
:perminfoindex 4
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname region
:colnames ("r_regionkey" "r_name" "r_comment")
}
}
:rtekind 0
:relid 17164
:inh false
:relkind r
:rellockmode 1
:perminfoindex 5
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname s
:colnames <>
}
}
:eref
{ALIAS
:aliasname s
:colnames ("s_supkey" "s_name" "s_address" "s_nationkey" "s_phone" "
s_acctbal" "s_comment")
}
}
:rtekind 0
:relid 17181
:inh false
:relkind r
:rellockmode 1
:perminfoindex 6
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname n
:colnames <>
}
}
:eref
{ALIAS
:aliasname n
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
}
:rtekind 0
:relid 17158
:inh false
:relkind r
:rellockmode 1
:perminfoindex 7
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname r
:colnames <>
}
}
:eref
{ALIAS
:aliasname r
:colnames ("r_regionkey" "r_name" "r_comment")
}
}
:rtekind 0
:relid 17164
:inh false
:relkind r
:rellockmode 1
:perminfoindex 8
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17169
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10 12 13)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17263
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17191
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17158
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17164
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17181
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12 13 14)
:insertedCols (b)

```

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:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17158
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17164
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 6
}
)
:quals (
{OEXPR
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 6
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 6
:location -1
}
{CONST
:consttype 1700
:consttypmod -1
:constcollid 0
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 6 [ 24 0 0 0 0 -128 ]
}
)
:location -1
}
)
}
:rarg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 7
}
{RANGETBLREF
:rtindex 8
}
)
:quals <>
}
:usingClause <>
:join_using_alias <>
:quals (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 7
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 7
:varattnosyn 1
:location -1
}
{VAR
:varno 6
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 4
:location -1
}
)
:location -1
}
)
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 7
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 7
:varattnosyn 3
:location -1
}
)
{VAR

```

```

:varno 8
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 8
:varattnosyn 1
:location -1
}
)
:location -1
}
{OPEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 8
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 8
:varattnosyn 2
:location -1
}
{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
}
)
:location -1
}
)
:alias <>
:rtindex 0
}
{RANGETBLREF
:rtindex 3
}
{RANGETBLREF
:rtindex 4
}
{RANGETBLREF
:rtindex 5
}
)
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 290
}
{VAR
:varno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 302
}
)
:location 300
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 6
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 1
:location -1
}
{VAR
:varno 3
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 2
:location 337
}
)
:location 335
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1

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```

:varattno 6
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 360
}
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 369
:constvalue 4 [ 15 0 0 0 0 0 0 0 ]
}
)
:location 367
}
{OEXPR
:opno 1209
:opfuncid 850
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{RELABELTYPE
:arg
{VAR
:varno 1
:varattno 5
:vartype 1043
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 384
}
:resulttype 25
:resulttypmod -1
:resultcollid 100
:relabelformat 2
:location -1
}
{CONST
:consttype 25
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 396
:constvalue 10 [ 40 0 0 0 0 37 66 82 65 83 83 ]
}
}
:location 391
}
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 6
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 4
:location -1
}
{VAR
:varno 4
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location 431
}
}
:location 429
}
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 4
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 3
:location 455
}
{VAR
:varno 5
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 1
:location 469
}
}
:location 467
}
{OEXPR
:opno 1054
:opfuncid 1040
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (

```

```

{VAR
:varno 5
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 2
:location 500
}
{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 516
:constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
}
}
:location 514
}
{OPEXPR
:opno 1752
:opfuncid 1718
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 4
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 4
:location 537
}
{SUBPLAN
:sublinktype 4
:testexpr <>
:paramids <>
:plan_id 1
:plan_name SubPlan\ 1
:firstcoltype 1700
:firstcoltypmod -1
:firstcolcollation 0
:usehashtable false
:unknowneqfalse false
:parallel_safe false
:setparam <>
:parparam (i 0)
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 826
}
}
)
:startup_cost 0
:per_call_cost 14030.867625000003
}
}
:location 551
}
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 6
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 6
:location -1
}
}
:resno 1
:resname s_acctbal
:ressortgroupref 1
:resorigtbl 17263
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 6
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 2
:location -1
}
}
:resno 2
:resname s_name
:ressortgroupref 3
:resorigtbl 17263
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 2
:location 86
}
}
:resno 3

```

```

:resname n_name
:ressortgroupref 2
:resorigtbl 17158
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 102
}
:resno 4
:resname p_partkey
:ressortgroupref 4
:resorigtbl 17169
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 121
}
:resno 5
:resname p_mfg
:ressortgroupref 0
:resorigtbl 17169
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 6
:varattno 3
:vartype 1043
:vartypmod 44
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 3
:location -1
}
:resno 6
:resname s_address
:ressortgroupref 0
:resorigtbl 17263
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 6
:varattno 5
:vartype 1042
:vartypmod 19
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 5
:location -1
}
:resno 7
:resname s_phone
:ressortgroupref 0
:resorigtbl 17263
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 6
:varattno 7
:vartype 1043
:vartypmod 105
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 7
:location -1
}
:resno 8
:resname s_comment
:ressortgroupref 0
:resorigtbl 17263
:resorigcol 7
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause {
{SORTGROUPCLAUSE
:tltSortGroupRef 1
:eqop 1752
:sortop 1756
:reverse_sort true
:nulls_first true
:hashable true
}
{SORTGROUPCLAUSE
:tltSortGroupRef 2
:eqop 1054
:sortop 1058
:reverse_sort false
:nulls_first false
:hashable true
}
{SORTGROUPCLAUSE
:tltSortGroupRef 3
:eqop 1054

```

```

:isortop 1058
:reverse_sort false
:nulls_first false
:hashable true

(SORTGROUPCLAUSE
 :tltSortGroupRef 4
 :eqop 96
 :isortop 97
 :reverse_sort false
 :nulls_first false
 :hashable true

)

:limitOffset <>
:limitCount
(CONST
 :consttype 20
 :consttyped -1
 :constcollid 0
 :constlen 8
 :constbyval true
 :constnull false
 :location -1
 :constval 8 [ 100 0 0 0 0 0 0 0 ]

)

:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
)

WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause {OPEXPR :opno 1209 :ofuncid 850 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ((RELABELTYPE :arg {VAR :varno 1 :varattno 5 :vartype 1043 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 5 :location -1} :resulttype 25 :resulttypmod -1 :resultcollid 100 :relabelformat 2 :location -1} (CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constval 10 [ 40 0 0 0 37 66 82 65 83 83 ])) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 8 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_select -1 :outer_select -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0} (RES TRICTINFO :clause {OPEXPR :opno 96 :ofuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ((VAR :varno 1 :varattno 6 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} (CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constval 4 [ 15 0 0 0 0 0 0 0 ])) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 7 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_select -1 :outer_select -1 :mergeopfamilies (b 1976) :left_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 6 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :j_d_relds (b 1 3 4 5 6 7 8)}) :right_em (EQUIVALENCEMEMBER :em_expr {CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constval 4 [ 15 0 0 0 0 0 0 0 ]} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :j_d_relds (b 1 3 4 5 6 7 8)}) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause {OPEXPR :opno 1209 :ofuncid 850 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ((RELABELTYPE :arg {VAR :varno 1 :varattno 5 :vartype 1043 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 5 :location -1} :resulttype 25 :resulttypmod -1 :resultcollid 100 :relabelformat 2 :location -1} (CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constval 10 [ 40 0 0 0 37 66 82 65 83 83 ])) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 8 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_select -1 :outer_select -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0} (RE STRICTINFO :clause {OPEXPR :opno 96 :ofuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ((VAR :varno 1 :varattno 6 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} (CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constval 4 [ 15 0 0 0 0 0 0 0 ])) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 7 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_select -1 :outer_select -1 :mergeopfamilies (b 1976) :left_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 6 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :j_d_relds (b 1 3 4 5 6 7 8)}) :right_em (EQUIVALENCEMEMBER :em_expr {CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constval 4 [ 15 0 0 0 0 0 0 0 ]} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :j_d_relds (b 1 3 4 5 6 7 8)}) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 3 :varattno 4 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 4 :location -1}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 3 :varattno 4 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 4 :location -1}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 4 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 2 :location -1} (VAR :varno 4 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 1 :location -1}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 4 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 2 :location -1} (VAR :varno 4 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 1 :location -1}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause {OPEXPR :opno 1054 :ofuncid 1048 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ((VAR :varno 5 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 5 :varattnosyn 2 :location -1} (CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constval 10 [ 40 0 0 0 69 85 82 79 80 69 ])) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 5) :required_relds (b 5) :incompatible_relds (b) :outer_relds (b 5) :right_relds (b) :orclause <> :rinfo_serial 14 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_select -1 :outer_select -1 :mergeopfamilies (b 426 2097) :left_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 5 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 5 :varattnosyn 2 :location -1} :em_relds (b 5) :em_is_const false :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :j_d_relds (b 1 3 4 5 6 7 8)}) :right_em (EQUIVALENCEMEMBER :em_expr {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constval 10 [ 40 0 0 0 69 85 82 79 80 69 ]} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :j_d_relds (b 1 3 4 5 6 7 8)}) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause {OPEXPR :opno 1054 :ofuncid 1048 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ((VAR :varno 5 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 5 :varattnosyn 2 :location -1} (CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constval 10 [ 40 0 0 0 69 85 82 79 80 69 ])) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 5) :required_relds (b 5) :incompatible_relds (b) :outer_relds (b 5) :right_relds (b) :orclause <> :rinfo_serial
```

```

operator 0 :left_bucketsize 1 :right_bucketsize 1 :left_mcvfreq 1 :right_mcvfreq 1 :left_hashoperator 0 :right_hashoperator 0)
WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause (OPEXPR :opno 1854 :opfuncid 10408 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ((VAR :varno 8 :varrattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 8 :varrattnosyn 2 :location -1) :constn 1 :constval false :constnull false :location -1) :constvalue 10 [ 4 0 0 0 6 9 8 5 8 2 7 9 8 0 6 9 ] :location -1) :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 8) :required_relds (b 8) :incompatible_relds (b) :outer_relds (b) :left_relds (b 8) :right_relds (b 3) :orclause > :rinfo_serial 15 :eval_cost_startup -1 :eval_cost_per_tuple 0 :norm_selc 0 :0058823529411764705 :outer_selc -1 :mergeofamilies (0 1976) :left_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 8 :varrattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 8 :varrattnosyn 2 :location -1) :em_relds (b 8) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 3 4 5 6 7 8)) :right_em (EQUIVALENCEMEMBER :em_expr (CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1) :constval false :constnull false :location -1) :constvalue 10 [ 4 0 0 0 6 9 8 5 8 2 7 9 8 0 6 9 ] :em_relds (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain (JOINDOMAIN :jd_relds (b 1 3 4 5 6 7 8)) :outer_is_left false :hashjoinoperator 0 :left_bucketsize 1 :right_bucketsize 1 :left_mcvfreq 1 :right_mcvfreq 1 :left_hashoperator 0 :right_hashoperator 0) | parallel_safe = true | reason = PROPAGALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 8 :varrattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 8 :varrattnosyn 1 :location -1) | parallel_safe = true | reason = PROPAGALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause (OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varrattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varrattnosyn 1 :location -1) (VAR :varno 3 :varrattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varrattnosyn 1 :location -1) :location -1) :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 1 3) :required_relds (b 1 3) :incompatible_relds (b) :outer_relds (b) :left_relds (b 3) :right_relds (b 3) :orclause > :rinfo_serial 15 :eval_cost_startup -1 :eval_cost_per_tuple 0 :norm_selc 1.197464381878494366e-05 :outer_selc -1 :mergeofamilies (0 1988) :left_em (EQUIVALENCEMEMBER :em_expr (SUBPLAN :sublinkType 4 :testexpr > :paramsid > :plan_id 1 :plan_name SubPlan\ 1 :firstCollType 1700 :firstCollation 0 :useHashTable false :unknownEqFalse false :parallel_safe false :setParam > :parParam (i 0) :args ((VAR :varno 1 :varrattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varrattnosyn 1 :location -1) :em_relds (b 3) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 3 4 5 6 7 8)) :right_em (EQUIVALENCEMEMBER :em_expr (CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen -1) :constval false :constnull false :location -1) :constvalue 1700 :em_relds (b 3) :em_is_const false :em_is_child false :em_datatype 1700 :em_jdomain (JOINDOMAIN :jd_relds (b 1 3 4 5 6 7 8)) :outer_is_left false :hashjoinoperator 1752 :left_bucketsize 1 :right_bucketsize 1 :left_mcvfreq 1 :right_mcvfreq 1 :left_hashoperator 1752 :right_hashoperator 1752) | parallel_safe = false | reason = PROPAGALLEL_RESTRICTED
WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause (OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 6 :varrattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varrattnosyn 1 :location -1) (VAR :varno 3 :varrattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varrattnosyn 1 :location -1) :location -1) :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 1 3) :required_relds (b 1 3) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b 3) :orclause > :rinfo_serial 15 :eval_cost_startup -1 :eval_cost_per_tuple 0 :norm_selc 1.197464381878494366e-05 :outer_selc -1 :mergeofamilies (0 1988) :left_em (EQUIVALENCEMEMBER :em_expr (SUBPLAN :sublinkType 4 :testexpr > :paramsid > :plan_id 1 :plan_name SubPlan\ 1 :firstCollType 1700 :firstCollation 0 :useHashTable false :unknownEqFalse false :parallel_safe false :setParam > :parParam (i 0) :args ((VAR :varno 1 :varrattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varrattnosyn 1 :location -1) :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 3 4 5 6 7 8)) :right_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 3 :varrattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varrattnosyn 1 :location -1) :em_relds (b 6) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 3 4 5 6 7 8)) :outer_is_left false :hashjoinoperator 96 :left_bucketsize 1 :right_bucketsize 1 :left_mcvfreq 1 :right_mcvfreq 1 :left_hashoperator 96 :right_hashoperator 96) (RESTRICTINFO :clause (OPEXPR :opno 1752 :opfuncid 1718 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 6 :varrattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varrattnosyn 1 :location -1) (VAR :varno 3 :varrattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varrattnosyn 1 :location -1) :location -1) :start_cost 14030.867625000003 :VAR :varno 3 :varrattno 4 :vartype 1700 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varrattnosyn 4 :location -1) :em_relds (b 3) :em_is_const false :em_is_child false :em_datatype 1700 :em_jdomain (JOINDOMAIN :jd_relds (b 1 3 4 5 6 7 8)) :outer_is_left false :hashjoinoperator 1752 :left_bucketsize 1 :right_bucketsize 1 :left_mcvfreq 1 :right_mcvfreq 1 :left_hashoperator 1752 :right_hashoperator 1752) | parallel_safe = false | reason = PROPAGALLEL_RESTRICTED
WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause (OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 6 :varrattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varrattnosyn 1 :location -1) (VAR :varno 3 :varrattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varrattnosyn 1 :location -1) :location -1) :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 3 6) :required_relds (b 3 6) :incompatible_relds (b) :outer_relds (b 6) :left_relds (b 6) :right_relds (b 3) :orclause > :rinfo_serial 17 :eval_cost_startup -1 :eval_cost_per_tuple 0 :norm_selc 0.000199960079984003 :outer_selc -1 :mergeofamilies (0 1976) :left_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 6 :varrattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varrattnosyn 1 :location -1) :em_relds (b 6) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 3 4 5 6 7 8)) :right_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 3 :varrattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varrattnosyn 2 :location -1) :em_relds (b 3) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 3 4 5 6 7 8)) :outer_is_left false :hashjoinoperator 96 :left_bucketsize 1 :right_bucketsize 1 :left_mcvfreq 1 :right_mcvfreq 1 :left_hashoperator 96 :right_hashoperator 96)
WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause (OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 6 :varrattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varrattnosyn 1 :location -1) (VAR :varno 3 :varrattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varrattnosyn 2 :location -1) :location -1) :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 3 6) :required_relds (b 3 6) :incompatible_relds (b) :outer_relds (b 6) :left_relds (b 6) :right_relds (b 3) :orclause > :rinfo_serial 17 :eval_cost_startup -1 :eval_cost_per_tuple 0 :norm_selc 0.000199960079984003 :outer_selc -1 :mergeofamilies (0 1976) :left_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 6 :varrattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevels
```

[illegible]

[illegible]

[illegible]


```

Buffers: shared hit=10484
Worker 0: actual time=6.763..47.240 rows=31790.00 loops=1
  Buffers: shared hit=4496
Worker 1: actual time=6.574..51.345 rows=34626.00 loops=1
  Buffers: shared hit=4856
-> Parallel Seq Scan on public.partsupp (cost=0.00..10458.67 rows=16667 width=14) (actual time=0.021..10.490 rows=13333.33 loops=3)
  Output: partsupp.ps_partkey, partsupp.ps_suppkey, partsupp.ps_availqty, partsupp.ps_supplycost, partsupp.ps_comment
  Buffers: shared hit=8792
  Worker 0: actual time=0.040..14.230 rows=178735.00 loops=1
    Buffers: shared hit=3932
  Worker 1: actual time=0.013..15.890 rows=195360.00 loops=1
    Buffers: shared hit=4292
-> Hash (cost=237.55..237.55 rows=1 width=245) (actual time=6.777..6.783 rows=887.00 loops=3)
  Output: s.s_acctbal, s.s_name, s.s_address, s.s_phone, s.s_comment, s.s_suppkey, nation.n_name
  Buckets: 1024 Batches: 1 Memory Usage: 184kB
  Buffers: shared hit=1692
  Worker 0: actual time=6.711..6.716 rows=887.00 loops=1
    Buffers: shared hit=564
  Worker 1: actual time=6.550..6.557 rows=887.00 loops=1
    Buffers: shared hit=564
-> Nested Loop (cost=24.50..237.55 rows=1 width=245) (actual time=0.163..6.477 rows=887.00 loops=3)
  Output: s.s_acctbal, s.s_name, s.s_address, s.s_phone, s.s_comment, s.s_suppkey, nation.n_name
  Join Filter: (nation.n_nationkey = n.n_nationkey)
  Rows Removed by Join Filter: 3548
  Buffers: shared hit=1692
  Worker 0: actual time=0.208..6.423 rows=887.00 loops=1
    Buffers: shared hit=564
  Worker 1: actual time=0.199..6.268 rows=887.00 loops=1
    Buffers: shared hit=564
-> Nested Loop (cost=0.00..25.95 rows=1 width=108) (actual time=0.079..0.086 rows=5.00 loops=3)
  Output: nation.n_name, nation.n_nationkey
  Join Filter: (nation.n_regionkey = region.r_regionkey)
  Rows Removed by Join Filter: 20
  Buffers: shared hit=6
  Worker 0: actual time=0.115..0.121 rows=5.00 loops=1
    Buffers: shared hit=2
  Worker 1: actual time=0.101..0.107 rows=5.00 loops=1
    Buffers: shared hit=2
-> Seq Scan on public.region (cost=0.00..12.12 rows=1 width=4) (actual time=0.048..0.050 rows=1.00 loops=3)
  Output: region.r_regionkey, region.r_name, region.r_comment
  Filter: (region.r_name = 'EUROPE'::bpchar)
  Rows Removed by Filter: 4
  Buffers: shared hit=3
  Worker 0: actual time=0.072..0.074 rows=1.00 loops=1
    Buffers: shared hit=1
  Worker 1: actual time=0.060..0.063 rows=1.00 loops=1
    Buffers: shared hit=1
-> Seq Scan on public.nation (cost=0.00..11.70 rows=170 width=112) (actual time=0.029..0.031 rows=25.00 loops=3)
  Output: nation.n_nationkey, nation.n_name, nation.n_regionkey, nation.n_comment
  Buffers: shared hit=3
  Worker 0: actual time=0.041..0.043 rows=25.00 loops=1
    Buffers: shared hit=1
  Worker 1: actual time=0.039..0.041 rows=25.00 loops=1
    Buffers: shared hit=1
-> Hash Semi Join (cost=24.50..211.26 rows=27 width=149) (actual time=0.015..1.225 rows=887.00 loops=15)
  Output: s.s_acctbal, s.s_name, s.s_address, s.s_phone, s.s_comment, s.s_nationkey, s.s_suppkey, n.n_nationkey
  Hash Cond: (s.s_nationkey = n.n_nationkey)
  Buffers: shared hit=1686
  Worker 0: actual time=0.016..1.210 rows=887.00 loops=5
    Buffers: shared hit=562
  Worker 1: actual time=0.017..1.181 rows=887.00 loops=5
    Buffers: shared hit=562
-> Seq Scan on public.supplier s (cost=0.00..174.50 rows=4558 width=145) (actual time=0.006..0.868 rows=4566.00 loops=15)
  Output: s.s_suppkey, s.s_name, s.s_address, s.s_nationkey, s.s_phone, s.s_acctbal, s.s_comment
  Filter: (s.s_acctbal > '0'::numeric)
  Rows Removed by Filter: 434
  Buffers: shared hit=1680
  Worker 0: actual time=0.008..0.862 rows=4566.00 loops=5
    Buffers: shared hit=500
  Worker 1: actual time=0.008..0.834 rows=4566.00 loops=5
    Buffers: shared hit=500
-> Hash (cost=24.48..24.48 rows=1 width=4) (actual time=0.029..0.031 rows=5.00 loops=3)
  Output: n.n_nationkey
  Buckets: 1024 Batches: 1 Memory Usage: 9kB
  Buffers: shared hit=6
  Worker 0: actual time=0.028..0.029 rows=5.00 loops=1
    Buffers: shared hit=2
  Worker 1: actual time=0.031..0.032 rows=5.00 loops=1
    Buffers: shared hit=2
-> Hash Join (cost=12.14..24.48 rows=1 width=4) (actual time=0.021..0.026 rows=5.00 loops=3)
  Output: n.n_nationkey
  Hash Cond: (n.n_regionkey = r.r_regionkey)
  Buffers: shared hit=6
  Worker 0: actual time=0.021..0.025 rows=5.00 loops=1
    Buffers: shared hit=2
  Worker 1: actual time=0.024..0.027 rows=5.00 loops=1
    Buffers: shared hit=2
-> Seq Scan on public.nation n (cost=0.00..11.70 rows=170 width=8) (actual time=0.002..0.004 rows=25.00 loops=3)
  Output: n.n_nationkey, n.n_name, n.n_regionkey, n.n_comment
  Buffers: shared hit=3
  Worker 0: actual time=0.002..0.003 rows=25.00 loops=1
    Buffers: shared hit=1
  Worker 1: actual time=0.002..0.004 rows=25.00 loops=1
    Buffers: shared hit=1
-> Hash (cost=12.12..12.12 rows=1 width=4) (actual time=0.011..0.011 rows=1.00 loops=3)
  Output: r.r_regionkey
  Buckets: 1024 Batches: 1 Memory Usage: 9kB
  Buffers: shared hit=3
  Worker 0: actual time=0.012..0.012 rows=1.00 loops=1
    Buffers: shared hit=1
  Worker 1: actual time=0.012..0.013 rows=1.00 loops=1
    Buffers: shared hit=1
-> Seq Scan on public.region r (cost=0.00..12.12 rows=1 width=4) (actual time=0.004..0.005 rows=1.00 loops=3)
  Output: r.r_regionkey
  Filter: (r.r_name = 'EUROPE'::bpchar)
  Rows Removed by Filter: 4
  Buffers: shared hit=3
  Worker 0: actual time=0.005..0.006 rows=1.00 loops=1
    Buffers: shared hit=1
  Worker 1: actual time=0.005..0.006 rows=1.00 loops=1
    Buffers: shared hit=1
-> Hash (cost=3580.00..3580.00 rows=466 width=30) (actual time=12091.653..12091.655 rows=190.00 loops=1)
  Output: part.p_partkey, part.p_mfgr
  Buckets: 1024 Batches: 1 Memory Usage: 20kB
  Buffers: shared hit=3225848
-> Seq Scan on public.part (cost=0.00..3580.00 rows=466 width=30) (actual time=0.070..21.800 rows=362.00 loops=1)
  Output: part.p_partkey, part.p_mfgr
  Filter: (((part.p_type)::text ~ '%BRASS'::text) AND (part.p_size = 15))
  Rows Removed by Filter: 99638
  Buffers: shared hit=2080
SubPlan 1
-> Aggregate (cost=14030.86..14030.87 rows=1 width=32) (actual time=33.426..33.426 rows=1.00 loops=552)
  Output: min(partsupp_1.ps_supplycost)
  Buffers: shared hit=4916046
-> Nested Loop (cost=24.50..14030.86 rows=1 width=6) (actual time=32.834..33.425 rows=0.94 loops=552)
  Output: partsupp_1.ps_supplycost
  Join Filter: (nation_1.n_regionkey = region_1.r_regionkey)
  Buffers: shared hit=4916046
-> Nested Loop (cost=24.50..14018.72 rows=1 width=10) (actual time=32.831..33.421 rows=0.94 loops=552)
  Output: partsupp_1.ps_supplycost, nation_1.n_regionkey
  Join Filter: (nation_1.n_nationkey = n_1.n_nationkey)
  Rows Removed by Join Filter: 23
  Buffers: shared hit=4915528
-> Nested Loop (cost=24.50..14004.89 rows=1 width=14) (actual time=32.821..33.409 rows=0.94 loops=552)
  Output: s_1.s_nationkey, n_1.n_nationkey, partsupp_1.ps_supplycost
  Join Filter: (partsupp_1.ps_suppkey = s_1.s_suppkey)
  Rows Removed by Join Filter: 3547
  Buffers: shared hit=4915010
-> Hash Semi Join (cost=24.50..211.26 rows=27 width=12) (actual time=0.003..1.195 rows=887.00 loops=552)
  Output: s_1.s_nationkey, s_1.s_suppkey, n_1.n_nationkey
  Hash Cond: (s_1.s_nationkey = n_1.n_nationkey)
  Buffers: shared hit=61826
-> Seq Scan on public.supplier s_1 (cost=0.00..174.50 rows=4558 width=8) (actual time=0.001..0.079 rows=4566.00 loops=552)
  Output: s_1.s_suppkey, s_1.s_name, s_1.s_address, s_1.s_nationkey, s_1.s_phone, s_1.s_acctbal, s_1.s_comment
  Filter: (s_1.s_acctbal > '0'::numeric)
  Rows Removed by Filter: 434
  Buffers: shared hit=61824
-> Hash (cost=24.48..24.48 rows=1 width=4) (actual time=0.031..0.032 rows=5.00 loops=1)

```

```

Output: n_1.n_nationkey
Buckets: 1024 Batches: 1 Memory Usage: 9kB
Buffers: shared hit=2
-> Hash Join (cost=12.14..24.48 rows=1 width=4) (actual time=0.023..0.027 rows=5.00 loops=1)
    Output: n_1.n_nationkey
    Hash Cond: (n_1.n_regionkey = r_1.r_regionkey)
    Buffers: shared hit=2
    -> Seq Scan on public.nation n_1 (cost=0.00..11.70 rows=170 width=8) (actual time=0.004..0.005 rows=25.00 loops=1)
        Output: n_1.n_nationkey, n_1.n_name, n_1.n_regionkey, n_1.n_comment
        Buffers: shared hit=1
    -> Hash (cost=12.12..12.12 rows=1 width=4) (actual time=0.010..0.010 rows=1.00 loops=1)
        Output: r_1.r_regionkey
        Buckets: 1024 Batches: 1 Memory Usage: 9kB
        Buffers: shared hit=1
        -> Seq Scan on public.region r_1 (cost=0.00..12.12 rows=1 width=4) (actual time=0.004..0.004 rows=1.00 loops=1)
            Output: r_1.r_regionkey
            Filter: (r_1.r_name = 'EUROPE'::bpchar)
            Rows Removed by Filter: 4
            Buffers: shared hit=1
-> Materialize (cost=0.00..13792.02 rows=4 width=10) (actual time=0.017..0.036 rows=4.00 loops=489624)
    Output: partsupp_1.ps_supplycost, partsupp_1.ps_suppkey
    Storage: Memory Maximum Storage: 17kB
    Buffers: shared hit=4853184
    -> Seq Scan on public.partsupp partsupp_1 (cost=0.00..13792.00 rows=4 width=10) (actual time=15.438..31.766 rows=4.00 loops=552)
        Output: partsupp_1.ps_supplycost, partsupp_1.ps_suppkey
        Filter: (part.p_partkey = partsupp_1.ps_partkey)
        Rows Removed by Filter: 399996
        Buffers: shared hit=4853184
-> Seq Scan on public.nation nation_1 (cost=0.00..11.70 rows=170 width=8) (actual time=0.001..0.002 rows=25.00 loops=518)
    Output: nation_1.n_nationkey, nation_1.n_name, nation_1.n_regionkey, nation_1.n_comment
    Buffers: shared hit=518
-> Seq Scan on public.region region_1 (cost=0.00..12.12 rows=1 width=4) (actual time=0.001..0.001 rows=1.00 loops=518)
    Output: region_1.r_regionkey, region_1.r_name, region_1.r_comment
    Filter: (region_1.r_name = 'EUROPE'::bpchar)
    Rows Removed by Filter: 4
    Buffers: shared hit=518

```

```

Planning:
  Buffers: shared hit=214
Planning Time: 21.062 ms
Execution Time: 18504.326 ms
(208 rows)

```

BENCHMARK SUMMARY

```

Runs           : 10
Average Planning : 21.390 ms
Average Execution : 18709.279 ms

```

```

Generated Files:
  run10_explain.txt
  summary.txt

```

adityadave@Adityas-MacBook-Air-3 Q2&Q21 %